

RUBEN CARRAZCO

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EDUCATION

Stanford University

Expected Graduation: Jun. 2027

B.S. in Electrical Engineering (Hardware and Software Track), GPA: 3.70/4.00

Stanford, CA

RELEVANT COURSEWORK

Deep Learning for CV Computer Organization & Systems VLSI Systems Digital Systems Architecture Circuit Design

EXPERIENCE

THG Health Capital – HeartPulse

Aug. 2026 – Present

Software Engineering Lead (Software Engineering Intern until Sep. 2026)

Remote

- Engineered core workflows for a **HIPAA-regulated** pediatric in-home nursing platform, including turning doctors' orders into timed tasks for nurses on shift, medication logging, nurse license and background verification, and new secure logins for doctors and family caregivers; delivered **14 features** and **278 tickets**, landing **1,286 commits** in two repositories.
- Owned **4 compliance features** closing **46 tracked requirements** for records custody, data retention, and care timing, backed by **255 test files** (~3,900 tests), roughly a third of the project's 11,500-test suite.

Stanford University

Sep. 2025 – Present

Teaching Assistant, CS106A/B

Stanford, CA

- Lead weekly Python/C++ discussion sections for Stanford's **900+** student introductory programming sequence, working directly with approximately **15 students** per section.
- Provide debugging-focused mentorship during section and office hours, helping students build problem-solving skills in programming and algorithms.

PROJECTS

Physics-Constrained Thermal Map Predictor | PyTorch

Jan. – Jun. 2026

- Designed a **physics-constrained U-Net** that predicts full-chip thermal heatmaps from floorplan and power-density images, adding the 2D heat equation as a differentiable loss term instead of relying on simulator calls.
- Achieved **0.895 K RMSE** and **0.841 hotspot IoU** on CircuitNet 2.0 layouts, an **11% improvement** over an MSE-only baseline, by training and comparing 12 models (3 architectures × 4 loss weightings) on Modal.

DNN Inference AI Hardware Accelerator | HLS, RTL, UVM, Synopsys DC, Cadence Innovus

Jan. – Mar. 2026

- Implemented a **16×16 systolic-array** DNN accelerator in HLS-to-RTL, achieving **85% MAC utilization** through loop pipelining, unrolling, and dataflow optimization.
- Drove the design through full **ASIC flow**, including synthesis, place-and-route, and signoff, using Synopsys Design Compiler and Cadence Innovus.
- Developed a **UVM testbench** for functional verification and validation across the design flow.

SIMD GEMM Hardware Accelerator | HLS, SystemVerilog

Sep. – Dec. 2025

- Designed and optimized a SIMD GEMM accelerator, achieving **60%+ improvement** in power-area-latency figure of merit through architecture and resource tradeoff exploration.
- Reduced multiplier count from **7 to 4** (17% area reduction), then scaled to **64 PEs** for **4× latency reduction** while maintaining timing closure.

MIPS 5-Stage Pipelined Processor | SystemVerilog, FPGA

Jan. – Mar. 2026

- Implemented a **5-stage pipelined MIPS processor** supporting **40+ instructions** with full hazard resolution (forwarding, stalling, branch handling), validated on FPGA using a Sobel edge-detection workload.

AC-DC Converter with Ripple Filtering | LTspice, Oscilloscope, Waveform Generator

Jan. 2025 – Feb. 2025

- Designed and implemented a **full-bridge AC-DC converter** with MOSFET-based regulation across a **0–22V adjustable output range**, validated across **3 load conditions** (100 Ω–10 kΩ), achieving **<50 mV** peak ripple throughout.
- Characterized circuit performance using **oscilloscopes** and waveform generators; validated transient response through iterative **LTspice simulations**.

TECHNICAL SKILLS

Machine Learning & Data: PyTorch, CNNs, U-Net, Encoder-Decoder Architectures, NumPy, Data Pipelines

Programming: Python, C/C++, SystemVerilog (RTL, UVM), HLS, Assembly (MIPS, x86-64)

Hardware & Architecture: RTL Microarchitecture, Hardware Accelerators, PPA Optimization, Static Timing Analysis

EDA & Tools: Synopsys (VCS, DC), Cadence Innovus, Xilinx Vivado, LTspice, Git, Oscilloscopes